

transport, yet include functionalities traditionally found in several separate instruments. Some handheld models combine the features of standard oscilloscopes with digital multimeters, which is essential for on-site diagnostics and field services. Despite its sophisticated capabilities, for instance, the PicoScope 6428E-D is notably compact, enhancing its portability.

Modern handheld oscilloscopes like the PicoScope 6428E-D combine the features of standard oscilloscopes with digital multimeters, making them ideal for on-site diagnostics and field services. Despite their robust features, these devices remain compact, optimising portability in space-constrained environments. They also support extensive memory depth and high sampling rates, which are necessary for capturing and analysing complex electronic signals.

These instruments support extensive memory depth and high sampling rates, which are crucial for capturing intricate electronic signal behaviours. New connectivity features, such as USB-C charging and LAN-based remote operations, align these tools with modern, networked work environments, while large displays and multi-touch gestures significantly enhance ease of use and control.

Modernising oscilloscope technology

Digital oscilloscope technology continues to evolve, with expansions in bandwidth and sampling rates meeting the demands of electronic systems operating at higher frequencies. The evolution of user interfaces and visualisation capabilities, including larger, high-resolution displays and touchscreen interfaces, facilitates quicker adjustments and navigation. Meanwhile, innovative visualisation technologies like 3D waveform displays simplify complex analyses.

The integration of sophisticated analytical tools directly within

Future trends and educational outreach

High-speed signal analysis. The industry is embracing oscilloscopes with higher bandwidth and sampling rates to meet the demands of advanced applications like LIDAR and high-energy physics. With true 12-bit resolution, these devices enhance signal detail, accelerate diagnostics, and improve accuracy.

Sampling precision. Advancements in sampling capabilities, powered by high-performance ASIC chipsets, reduce quantisation errors, and improve precision. New visualisation modes—density, waterfall, perspective, and mosaic—provide deep insights into signal behaviours, which are critical for communication technologies like 100-800G and 1.6T Ethernet.

Educational initiatives. Oscilloscope technology is evolving beyond hardware advancements. Industry leaders are driving educational initiatives through expert presentations and workshops, equipping professionals with advanced measurement and testing skills to remain at the forefront of innovation.

oscilloscopes enables complex analyses to be performed on the device itself, streamlining processes and reducing the need for additional software. Future probes are expected to offer higher bandwidth and reduced loading effects, enhancing measurements in high-speed digital and RF applications.

For example, the Rohde & Schwarz MXO 5C series showcases modern oscilloscope technology by combining high-speed data acquisition, flexible connectivity for external displays, and an efficient, compact design. These features address the needs of today's electronic testing environments and indicate future trends in integrating advanced analysis and visualisation tools directly into oscilloscopes, enhancing their capabilities and versatility.

Upgraded displays and interfaces

Future digital oscilloscopes are set to offer advanced user interfaces coupled with enhanced visualisation capabilities. These devices will feature high-resolution displays with wider screens and more accurate colour representation, making it easier for engineers to view and analyse waveforms. The integration of touchscreen interfaces will simplify interactions, allowing users to control the device through intuitive gestures

and navigate through settings and tools more efficiently.

Emerging visualisation technologies are also rising. The inclusion of 3D waveform displays and user-defined visual layouts will further simplify the analysis of complex waveforms. These technologies aim to enhance the user experience by making the devices more accessible and easier to handle, even during intricate analysis tasks.

The Micsig MHO3 series oscilloscopes feature a 35.56cm (14-inch) touchscreen with a resolution of 1920 x 1200. This display supports touch operations, allowing users to interact with the oscilloscope efficiently. The user interface is designed for quick learning and operation, and the time base matrix display enables instant access to any time base setting, enhancing efficiency and usability.

Advances in probe technology and connectivity

Probe technology is crucial for ensuring the precision of waveform measurements. Upcoming developments in digital oscilloscopes will introduce probes with higher bandwidth and reduced loading effects, enabling more accurate signal measurements. Active probes, equipped with amplifiers and equalisation mechanisms, will be particularly beneficial for high-speed digital and